

BRIEF RESEARCH PROPOSAL · SEPTEMBER 2026

Personalized practice for attention to faces

Two exploratory approaches for autistic children and adults

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Preliminary research concept; no clinical approval or institutional affiliation implied

Purpose

Explore whether a short, personally meaningful interaction can support voluntary, comfortable attention to a speaker's face and comprehension of the message. Longer face-looking alone is not a measure of improvement, and eye contact should never be imposed.

Approach 1 · Interest-led speaking face

A brief interview using text, speech or picture choices identifies topics of interest; the participant confirms the final choice. A speaking face with selectable appearance and voice presents a short, age-respectful narrative matched to communication needs. At planned pauses, a symbol or simple question appears. The response may be a touch or picture choice; number reading is not assumed.

Prompt exposure and response periods are logged separately. Occasional prompt-free segments assess whether attention to the face and understanding also change outside the guided task. The participant may pause or stop at any time.

Approach 2 · Gradual shape-to-face transition

A simple shape gradually acquires facial features. For adults for whom number reading is suitable, a random number first appears outside the face; the participant enters it on a keyboard or selects it from four choices. With correct responses and continued comfort, the number moves step by step closer to the face and eventually into the face region without covering key facial features. The shape becomes more face-like in parallel. Difficulty or discomfort prompts an easier stage or a nonnumeric symbol. Looking at the number and unprompted attention to the face are measured separately: bringing a number closer does not itself demonstrate improved social attention.

This is a separate exploratory arm; a simplified or partial face cannot be assumed to feel easier for every participant.

Suggested starting point: pilot Approach 1 on a small scale. Consider Approach 2 separately after clinical review.

Individual adaptation

Child and adult versions should differ in presentation and language. Communication ability and response mode, rather than age alone, determine task difficulty. Comfort, assent and the right to stop are part of the study outcomes.

Technical feasibility and limits

Our initial review supports evaluating a gaze-specific pretrained model with individual calibration using ordinary cameras. In two sessions with one webcam and one person, a local L2CS-Net prototype recorded median errors of approximately 150–164 screen pixels. These observations are specific to that setup and do not establish clinical validity.

A first outcome should distinguish looking **inside a large face region, outside it, or uncertain**. The current evidence does not justify reliable eye-versus-mouth classification. Detecting a face or iris alone does not locate gaze on a screen. There is no single mobile-camera accuracy figure; devices and intended users require separate validation. Earlier mobile studies reported results that depended on calibration, device and face layout [1,2].

Proposed feasibility pilot

1. Agree participant groups, stimuli, session length and stopping rules.
2. Validate each camera and calibration on held-out targets; measure error, invalid-data rate, latency, lighting and moderate head movement.
3. Run short supervised sessions; assess comfort, comprehension and face attention during prompt-free segments, and review within-person trends.

Questions for your guidance

1. Should the first study recruit children, adults or separate groups? What communication and response criteria matter?
2. What should the primary outcome be: face-region attention, comprehension, comfort, or a combination? What change is meaningful?
3. Is a filmed person, prerecorded video or avatar preferable? Which content and visual prompts are acceptable?
4. What stopping and recovery rules should apply to discomfort, fatigue or refusal?
5. Can a subset be compared with a research-grade eye tracker? What measurement quality is needed for interpretation?
6. How should consent, child assent, data retention and any need for raw video be handled?

Selected references

[1] Krafska et al., [Eye Tracking for Everyone](#), CVPR 2016. [2] Strobl et al., [Look me in the eye](#), 2019. [3] Park et al., [Few-Shot Adaptive Gaze Estimation](#), ICCV 2019. [4] Hadjikhani et al., [Look me in the eyes](#), 2017. [5] [L2CS-Net](#).

The proposed methods and decisions are design recommendations, not established findings from these papers. Clinical review, device validation and ethics approval are required before participant research.